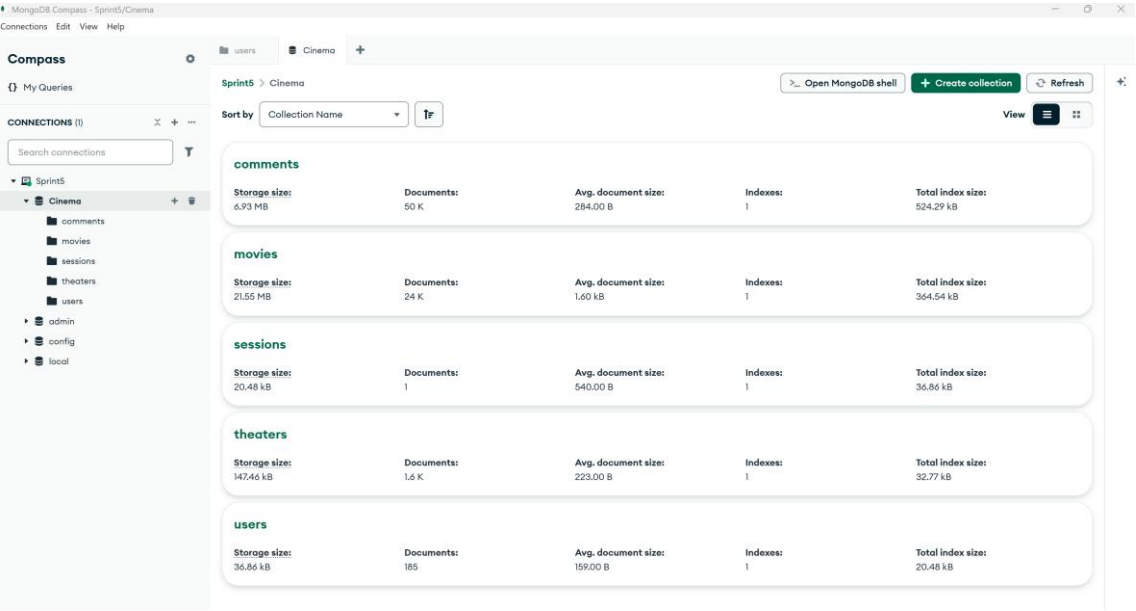


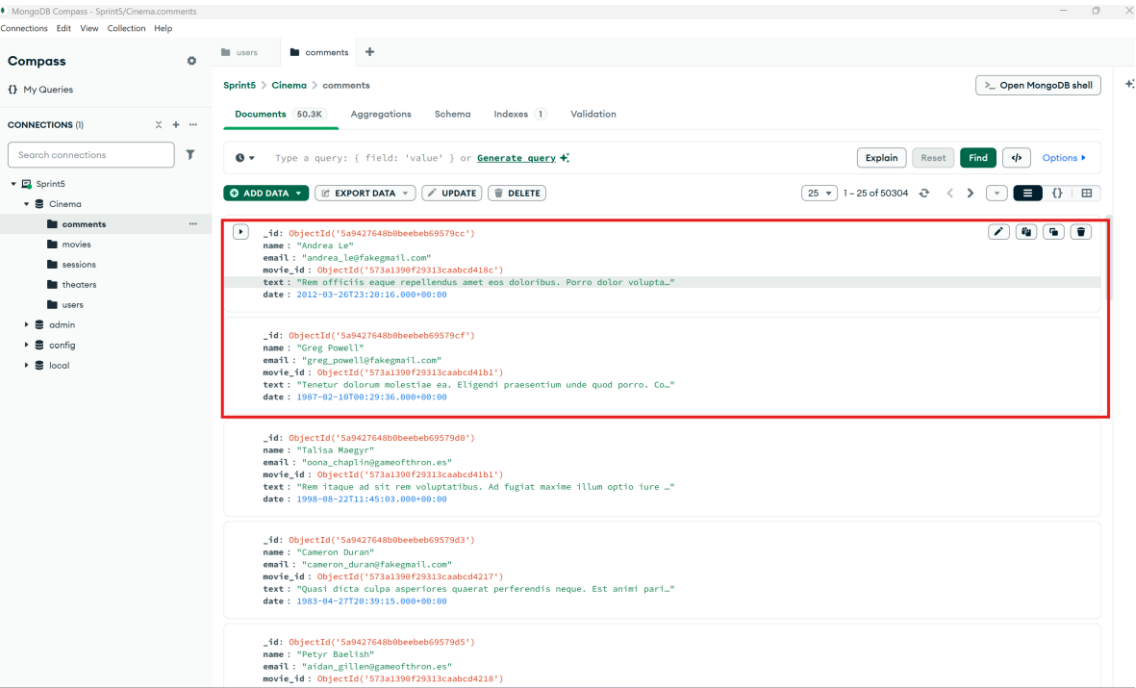
NIVELL 1:

Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.



EXERCICI 1:

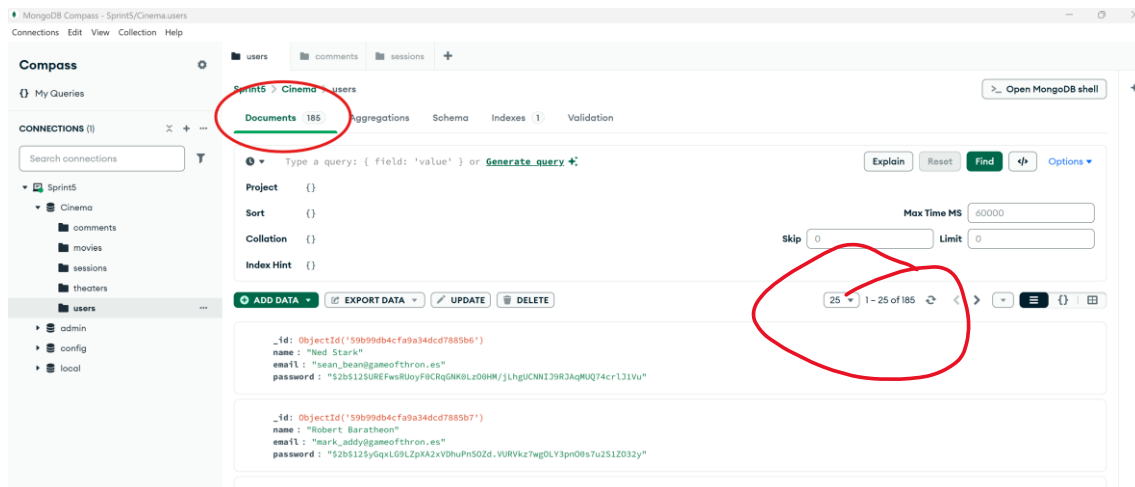
Mostra els 2 primers comentaris que hi ha en la base de dades.



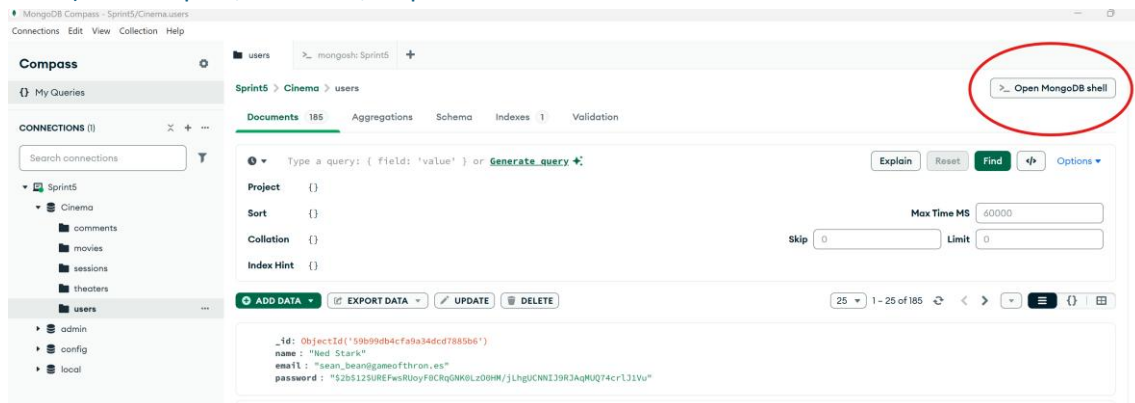
- **Quants usuaris tenim registrats?**

Per veure número de registres totals d'una col·lecció:

- Mirar el número de registres a la pestanya "documents". Si deixes el filtre buit, el resultat et diu el número.

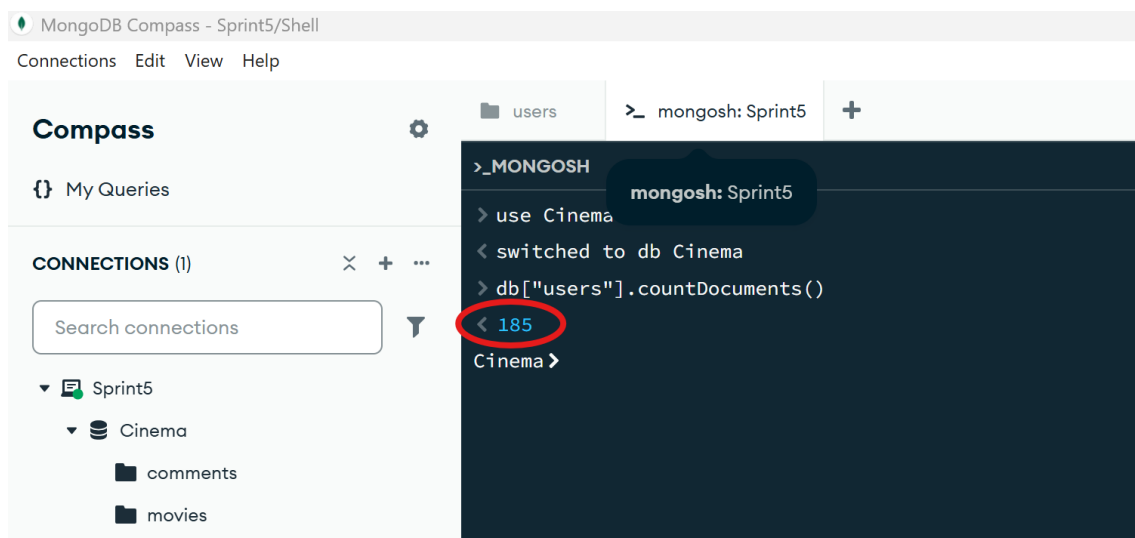


- A Compass, a la dreta, cliquem OPEN MONGODB SHELL



Escrivim el codi `db["users"].countDocuments()`

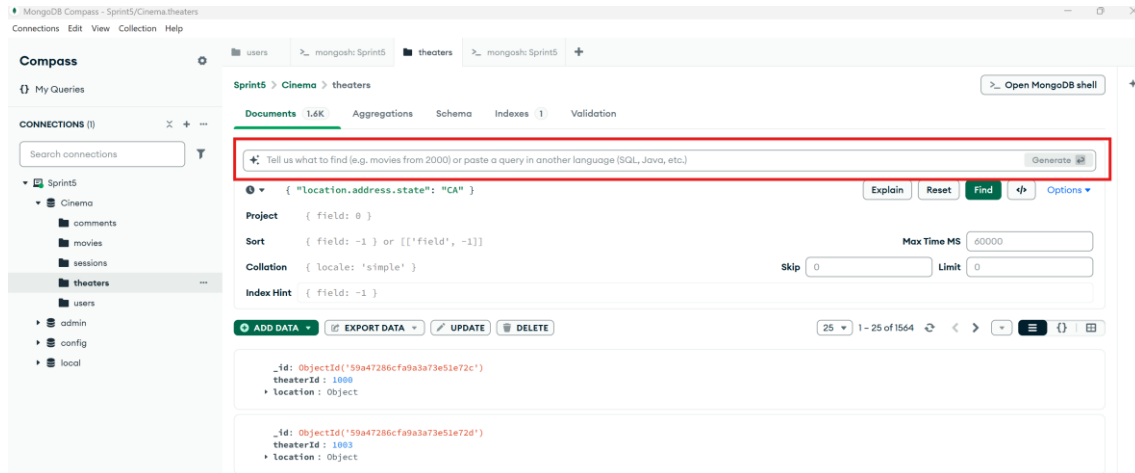
Més info a: [db.collection.countDocuments\(\) \(mongosh method\) - Database Manual - MongoDB](#)



o Quants cinemes hi ha en l'estat de Califòrnia?

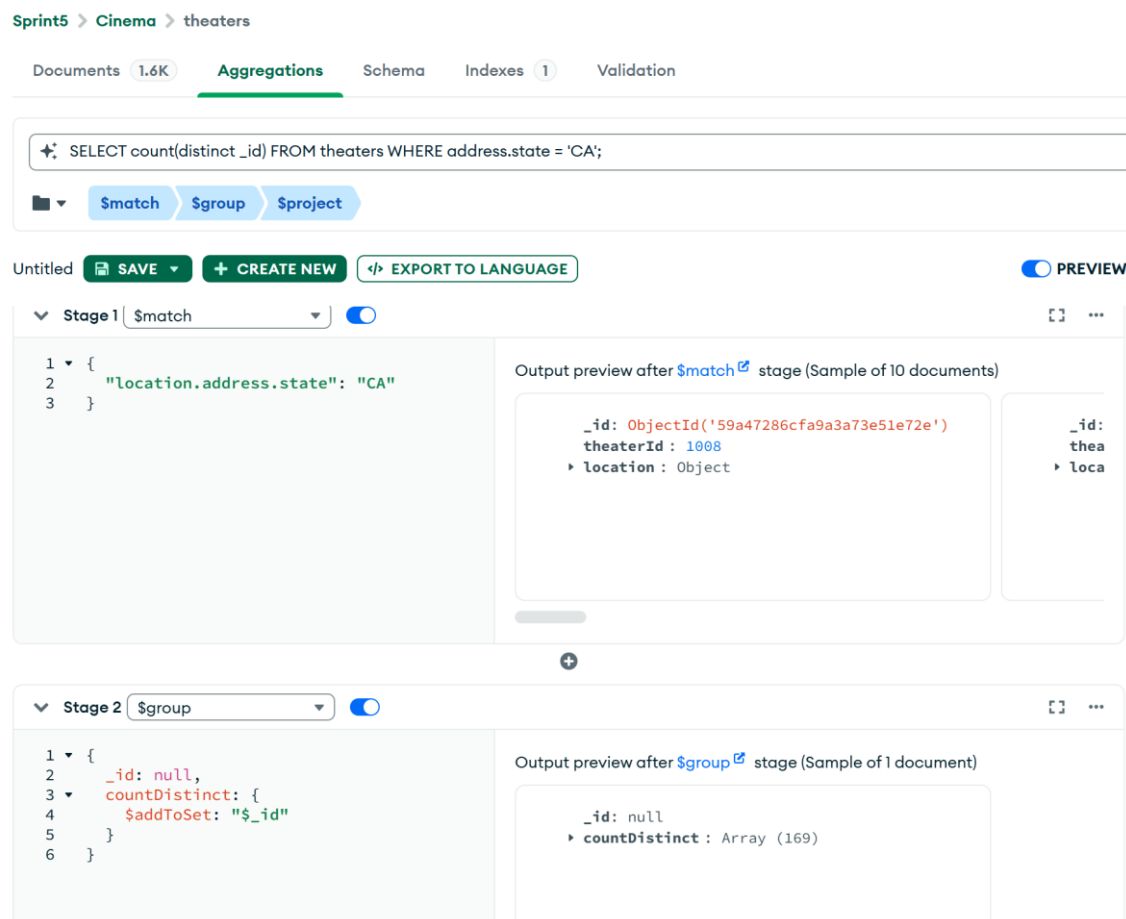
Opcions:

- a) A “Documents” – a la barra de codi escribim en SQL la consulta i cliquem Genera AI.
 “SELECT count(distinct _id) FROM theaters WHERE address.state = 'CA';”



Indicarà que el codi necessita agregacions i que es farà en pipeline.

El codi es dividirà en tres stages:



SELECT count(distinct _id) FROM theaters WHERE address.state = 'CA';

Buttons: \$match, \$group, \$project, Explain, Export, **Run**

Stage 3: \$project

Output preview after \$project stage (Sample of 1 document)

```
{
  "_id": null,
  "count": 169
}
```

Stage Wizard options:

- Find all the documents that match one or more conditions **\$MATCH**
- Group my documents based on field values **\$GROUP**
- Compute values within the group and create new fields **\$GROUP**
- Return a subset of values based on their order or rank **\$GROUP**
- Include or exclude a subset of fields from my documents **\$PROJECT**
- Sort documents based on a single field **\$SORT**

Al clicar run s'obté RESULTAT:

Sprint5 > Cinema > theaters

Buttons: \$match, \$group, \$project, Edit, Explain, Export, **Run**, Options

ALL RESULTS

Showing 1 - 1 count results

```
{
  "_id": null,
  "count": 169
}
```

Quin va ser el primer usuari/ària en registrar-se?

Escrivim el codi SQL i cliquem GENERATE AI.

MongoDB Compass - Sprint5/Cinema/users

Query: SELECT * FROM users ORDER BY _id DESC LIMIT 1;

Buttons: Generate AI, Explain, Reset, Find, Options

Results:

```
{
  "_id": ObjectId("5db1c37e4e68c31f18cf8a9f"),
  "name": "foo",
  "email": "foobaz@bar.com",
  "password": "foobar",
  "preferences": {}
}
```

o Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

Escrivim el codi en SQL. Mirem SCHEMA de la col·lecció movies per saber els camps i especialment conèixer com filtrar per comèdia. Camp: Genre. Pot ser string o array, pel què farem IN "Comedy":
 SELECT * FROM movies WHERE genres LIKE "Comedy"; Codi per fer find: {"genres": "Comedy"}. Com que és un filtre i després una agregació (recompte), es passa la generació de codi amb AI a una aggregation pipeline. Resultat: 7024

The screenshot shows the MongoDB Aggregations interface. At the top, the SQL query is: `SELECT count(*) FROM movies WHERE genres LIKE "Comedy";`. Below this, the aggregation pipeline is defined with two stages:

- Stage 1: \$match**

```
1 {
2   genres: "Comedy"
3 }
```

Output preview after \$match stage (Sample of 10 documents):

```
{
  "_id": ObjectId('573a1390f29313caabcd4803'),
  "plot": "Cartoon figures announce, via comic strip balloons, that they will mov...",
  "genres": Array (3),
  "runtime": 7,
  "cast": Array (1),
  "num_mflix_comments": 1,
  "poster": "https://m.media-amazon.com/images/M/MV5BYzgZnNjNhNT..."
}
```
- Stage 2: \$count**

```
1 "total"
```

Output preview after \$count stage (Sample of 1 document):

```
{
  "total": 7024
}
```

The value `7024` is circled in red in the output preview.

On the right, the 'Stage Wizard' panel lists various aggregation stages like \$MATCH, \$GROUP, \$SORT, etc.

EXERCICI 2. Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

Escrivim el codi en SQL i clico generate AI, que em transforma el codi a:

```
{"$and": [{"year": 1932}, {"$or": [{"genres": "Drama"}, {"languages": "French"}]}
```

The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS' panel lists the 'Sprint5' database and the 'Cinema' collection. The main panel displays a query for the 'movies' collection. The query is: `SELECT * FROM movies WHERE EXTRACT(year from produced) = 1932 AND (genres = 'Drama' OR languages = 'French')`. The results show a single document for the movie 'The Blood of a Poet' (1932), which is a French Drama. The document includes details like runtime, cast, poster, and awards.

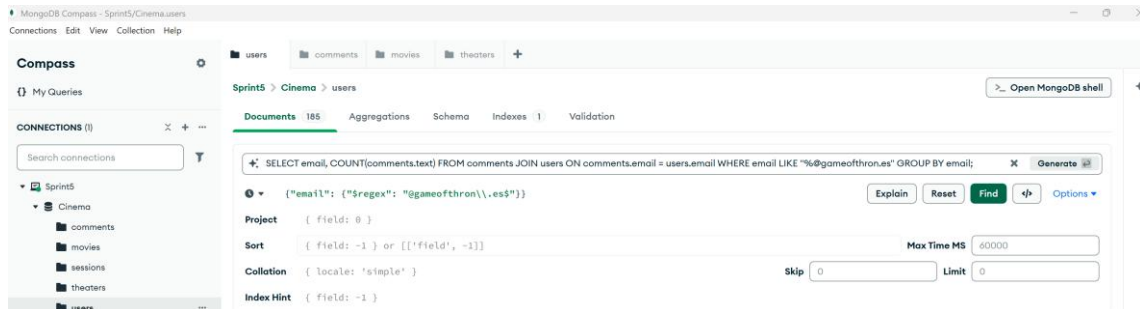
EXERCICI 3. Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

The screenshot shows the MongoDB Compass interface with a new query: `SELECT * FROM movies WHERE country = 'USA' AND wins BETWEEN 5 AND 9 AND year from produced BETWEEN 2012 AND 2014;`. The results show two documents. The first document is for the movie 'The Secret Life of Walter Mitty' (2013), which is a US production with 5 awards. The second document is for the movie 'After their cave is destroyed, a caveman family must trek through an u...' (2014), which is also a US production with 9 awards. The results panel shows 1-25 of 166 documents.

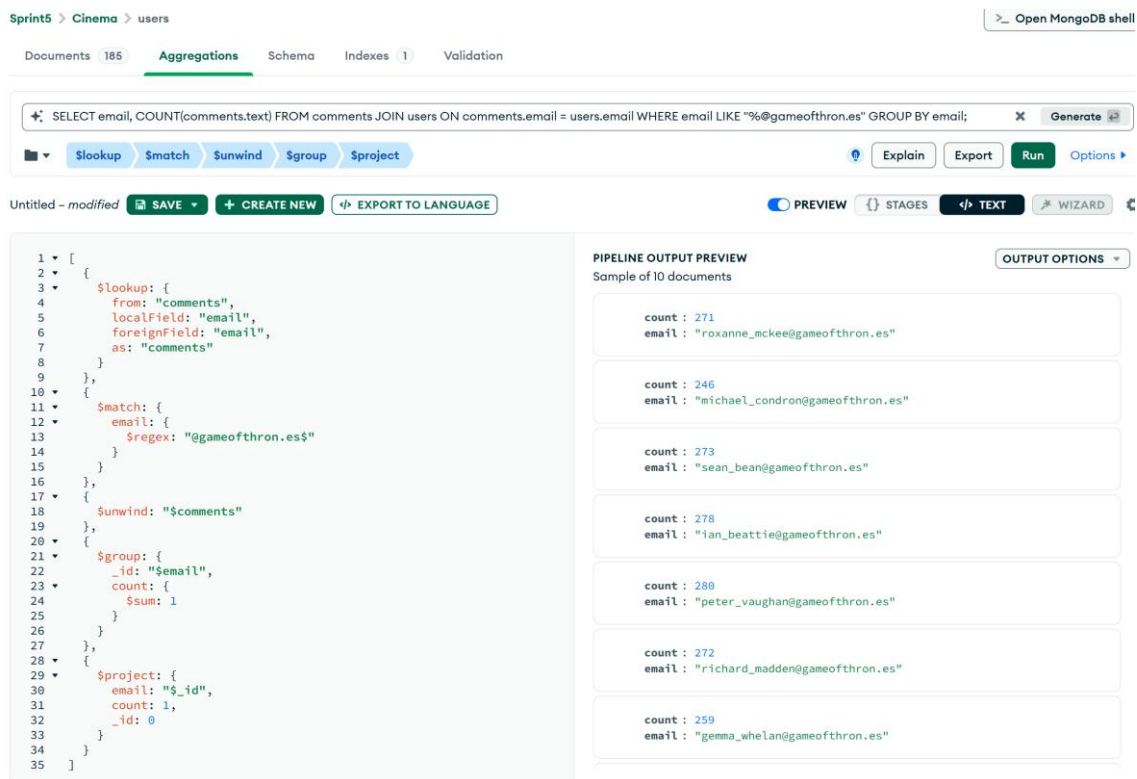
NIVELL 2:

EXERCICI 1: Compte quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

CODI SQL. Generate AI



Vista de text després de fer RUN a l'aggregation pipeline que es crea al generar codi amb AI.



EXERCICI 2. Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

✕ Generate

```
SELECT count(distinct theaterId), address.zipcode
FROM theaters
WHERE address.state = 'DC'
GROUP BY address.zipcode;
```

Smatch \$group Sproject

[Explain](#) [Export](#) [Run](#) [Options ▶](#)

PIPELINE OUTPUT PREVIEW

Sample of 3 documents

```
{
  "_id": "20002",
  "zipcode": "20002",
  "theaterIdCount": 1
}
```

```
{
  "_id": "20010",
  "zipcode": "20010",
  "theaterIdCount": 1
}
```

```
{
  "_id": "20016",
  "zipcode": "20016",
  "theaterIdCount": 1
}
```


NIVELL 3:

EXERCICI 1. Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.

```
SELECT * FROM movies WHERE directors LIKE "%JOHN LANDIS%" AND IMDB.rating BETWEEN 7,5 AND 8;

=

{"directors": { "$regex": "JOHN LANDIS", "$options": "i" }, "imdb.rating": { "$gte": 7.5, "$lte": 8 } }
```

Sprint5 > Cinema > movies

Documents 23.5K Aggregations Schema Indexes 1 Validation

SELECT * FROM movies WHERE directors LIKE "%JOHN LANDIS%" AND **IMDB**.rating BETWEEN 7,5 AND 8;

{ "directors": { "\$regex": "JOHN LANDIS", "\$options": "i" }, "**imdb**.rating": { "\$gte": 7.5, "\$lte": 8 } }

Generate

Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1-4 of 4

movies	_id ObjectId	fullplot String	imdb Object	year Int32	plot String	genres Array
1	ObjectId('573a1397f29313c...	"Faber College has one fr...	{ } 3 fields	1978	"At a 1962 college, Dean ...	[] 1 elemen
2	ObjectId('573a1397f29313c...	"After the release of Jak...	{ } 3 fields	1980	"Jake Blues, just out fro...	[] 3 elemen
3	ObjectId('573a1397f29313c...	"Two American college stu...	{ } 3 fields	1981	"Two American college stu...	[] 2 elemen
4	ObjectId('573a1398f29313c...	"Louis Winthorpe is a bus...	{ } 3 fields	1983	"A snobbish investor and ...	[] 1 elemen

Sprint5 > Cinema > movies

Documents 23.5K Aggregations Schema Indexes 1 Validation

SELECT * FROM movies WHERE directors LIKE "%JOHN LANDIS%" AND **IMDB**.rating BETWEEN 7,5 AND 8;

{ "directors": { "\$regex": "JOHN LANDIS", "\$options": "i" }, "**imdb**.rating": { "\$gte": 7.5, "\$lte": 8 } }

Generate

Options

ADD DATA EXPORT DATA UPDATE DELETE

25 1-4 of 4

_id: ObjectId('573a1397f29313caabce6d94')

fullplot: "Faber College has one frat house so disreputable it will take anyone. ..."

imdb: Object

year: 1978

plot: "At a 1962 college, Dean Vernon Wormer is determined to expel the entir..."

genres: Array (1)

rated: "R"

metacritic: 82

title: "Animal House"

lastupdated: "2015-09-13 00:02:47.883000000"

languages: Array (2)

writers: Array (3)

type: "movie"

tomatoes: Object

poster: "https://m.media-amazon.com/images/M/MV5BM2M2ZDA4MTYtOGRjWi00OTg5LWI1ZT..."

num_mflix_comments: 1

released: 1978-07-28T00:00:00.000+00:00

awards: Object

countries: Array (1)

cast: Array (4)

directors: Array (1)

runtime: 109

_id: ObjectId('573a1397f29313caabce76f7')

plot: "Jake Blues, just out from prison, puts together his old band to save t..."

genres: Array (3)

runtime: 133

rated: "R"

cast: Array (4)

num_mflix_comments: 1

poster: "https://m.media-amazon.com/images/M/MV5BYTdlMDExOGUxNzI3MS00MjY5LWE1NT..."

title: "The Blues Brothers"

fullplot: "After the release of Jake Blues from prison, he and brother Elwood go ..."

EXERCICI 2. Mostra en un mapa la ubicació de tots els teatres de la base de dades.

Tot i que sembla que les dades de coordenates lat i lon estan ben entrades, no apareix cap icona “MAP” al menú després de SCHEMA per poder veure les dades al mapa.

Mostro el mapa fet amb 1000 registres a través de la previsualització de coordenates a SCHEMA

Sprint5 > Cinema > theaters imdb

Documents 1.6K Aggregations **Schema** Indexes 1 Validation

✦ Tell us what to find (e.g. movies from 2000) or paste a query in another language (SQL, Java, etc.) Generate

Type a query: { field: 'value' } Reset Analyze </> Options

Project { field: 0 }

Sort { field: -1 } or [['field', -1]] Max Time MS 60000

Collation { locale: 'simple' } Skip 0 Limit 1564

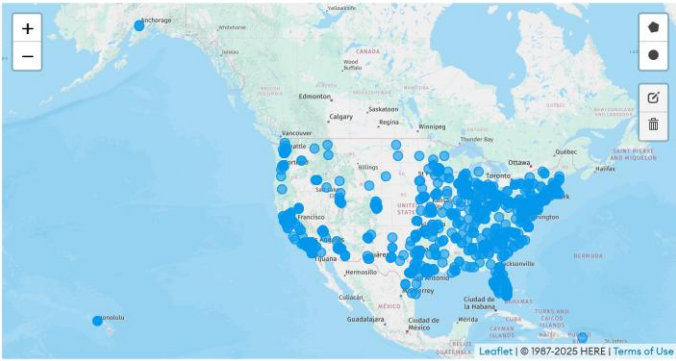
Index Hint { field: -1 }

[EXPORT SCHEMA](#) This report is based on a sample of 1000 documents. [Learn more](#)

document

geo

coordinates



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